



Data Driven Education

Discussion Forum Subteam

Presentation 1 - 02/14/2024

Introduction

- Discussion forums
 - How can we identify trends and unique data in discussion forum posts on MOOCs and Canvas/EdDiscussion platforms to help teachers leverage discussion forums better?
- How to analyze Student interaction
 - Previous models - Machine Learning team
 - Aspect-Based Sentiment Analysis model, Seeded Topic Modeling, Integrated LLMs
- Activity on forums indicative of student engagement and interest
 - Looking to consolidate, organize, and better display our work, finding new metrics to identify and looking for patterns amongst them.

Goal: Further the applications of our dashboards with variety of classes and different analysis

Recap of Fall 2023 and Next Steps

Category	Accomplished	Next Goals
Web Dev	• Completed two dashboards. • Researched Ed Discussion API • Interviewed TAs to get feedback on dashboards	• Implement TA feedback into dashboards • Make dashboards more robust and interactive with data science team findings
Data Science	• Seeded Topic Modeling with each course data • Metacognitive Presence Visualization • Data analytics on discussion post interactions and effectiveness	• Focus on cleaning and preparing data into a better format so that data analysis is easier and more streamlined • Look into more important metrics that we can identify

Focus for this semester

• Development

- Add another dashboard option?
 - Different accounts?
- Clean and prep dataset
 - Determine how to train model
- Visualizations for Dashboard

• Project Management

- Integrating option for a single teacher to view multiple dashboards
- Ensuring smooth communication and cooperation between Data Science to Dashboard pipeline
- Give tools for research on DS and WebDev
 - Supplement research done by previous subteam members by asking more questions with a specific focus on the tools that we are developing

• Research

- Determine if what model lead to helpful data for classrooms
- Focus interviews to the specific tools that we are developing and new classes data

- Student participation
 - Calculate a participation score in Piazza (or on our dashboard), without having to export the data manually (CS 2340 TA, Dr. Joyner, Spring 2021)
 - Report students with the highest participation, based on CP, endorsed-answers, and activity
 - Top student answers and top student “instructor-endorsed” answers (CS 1332 TAs, Spring 2021 + general, Fall 2021)
 - Report a metric of overall participation in the class
- Student sentiment
 - Overall sentiment of students (CS 2340 TA's, Spring 2021), although Vaibhav said that this largely stays the same semester to semester
 - Sentiment around a given HW or exam

Research Questions



- How can we design a web app that can provide actionable insights to the instructor?
 - how can we make the web app user-friendly and compatible with different devices?
- Which data science model/metrics would allow us to extract the best information from online discussion forums to establish a relationship between students and teachers?
- How can data science metrics be optimized to ensure seamless integration of information?

Data Science

Previous Goals

Goals Summary:

- **Data Cleaning:** Executed SQL queries to select relevant posts from the Piazza database, removing unnecessary data and preparing it for use by both the Data Science and Machine Learning teams.
- **Data Analysis:** Conducted analyses focusing on student-teacher interactions, such as response times and the quality of responses.

Key Achievements:

- Calculated the average response time between student posts and TA/instructor responses.
- Analyzed the number of posts per month and week to observe posting trends throughout the semester.
- Identified the number of posts made by instructors and students, as well as how many posts were endorsed by instructors.
- Prepared results and visualizations from Piazza data into a JSON file for the Web Development team.

Visual Elements:

- Diagrams or graphics illustrating the data cleaning and analysis processes.
- Charts or graphs displaying key metrics such as average response times, posting trends, and the number of endorsed posts.

Goals for the Semester

Enhance Data Metrics Utilization:

- Utilize metrics such as response time, unresolved posts, and total posts to provide educators with more comprehensive data visualizations.
- Use these metrics to allow teachers to adjust their teaching methods and for students to assess their engagement through discussion forums.

Deepen Data Analysis:

- Cross-examine metrics with the data file to uncover deeper relationships and more relevant applications of the data.
- Analyze student-teacher interaction data to improve educational outcomes and engagement.

Refine Data Cleaning Process:

- Continue to clean up data files, using tools like SQL, pandas, and Python, to ensure the data accurately reflects the metrics of interest.

Enhance Data Presentation:

- Create and refine a JSON file containing all relevant data, which will be integrated into the web development team's dashboard for real-time data representation and usage.

Programming and Analytics:

- Leverage high-level programming languages to explore and establish relationships between various data points related to student-teacher interactions.

Metrics/Data We're Using/ Why this Data is Relevant

Response time, Unresolved posts, total posts etc.

- These metrics help educators visualize the quantity and complexity of questions posed in class.
- Armed with this data, teachers can adjust their teaching approach or address specific questions during class to enhance student comprehension of course material.
- Additionally, students can view metrics of their participation in discussion forums, including the depth of their contributions aiming to enhance student engagement.



Steps To Take This Semester:

1. Look upon past work as a helpful guide to get started
2. Have a list of metrics prepared
3. Cross-examine our metrics with the data file to find relationships and relevant usage of the data
4. Clean up the data file we are using in a way that highlights our metrics
 - a. SQL, pandas, python
5. Collaborate with web dev team and other data scientists to find the most accurate and relevant data to display on the dashboard
6. Use any high level language program to find relationships between student-teacher interactions
7. Collaborate with other data science members to ensure your work is accurate
8. Create JSON file with all relevant data and send to web development team, who will use this information on their dashboard

Web Development

Previous Work

[Python Course](#)[AI Course](#)

Discussion Forum Analysis



Collect and analyze discussion posts from Piazza or Ed

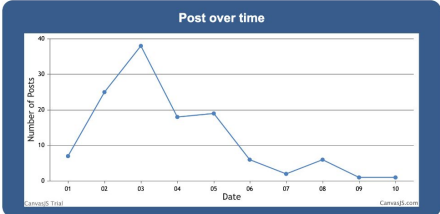


Display statistics on student and instructor participation and topics



Provide information and insight to improve student learning

Discussion Forum Analysis - AI Course

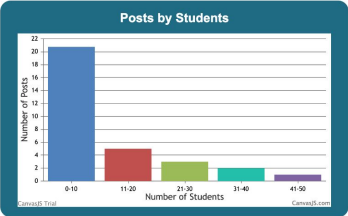


Current class sentiment:
0.4

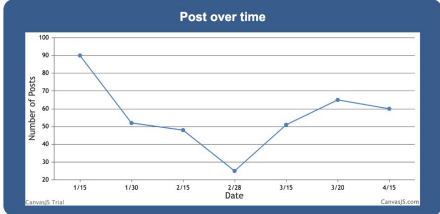
Total posts this week:
28

Average Score:
92.381

Customized Scores	
CP code	Frequency
0	814
1	210
2	164
3	159
4	146



Discussion Forum Analysis - Python Course

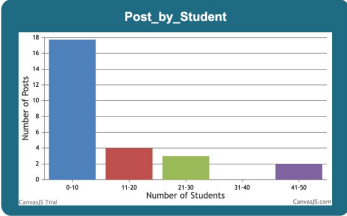


Current class sentiment:
0.44

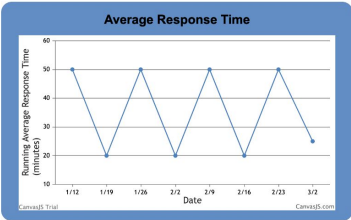
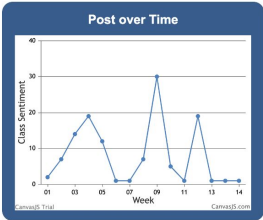
Total posts this week:
32

Average Score:
0.631

Customized Scores	
CP code	Frequency
0	814
1	210
2	164
3	159
4	146



Discussion Forum Analysis - AI Course



Student Teacher Topics

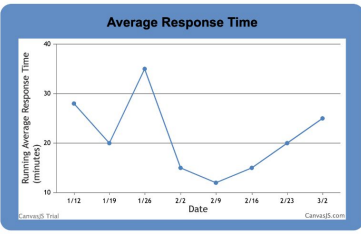
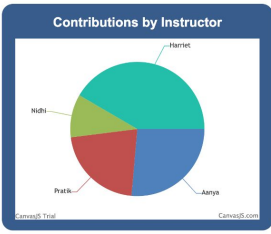
Current unresolved posts: 7

Today's average reply time (mins): 48

Current unresolved posts: 2

Unresolved Posts				
Post	Time	Subject	Question	Assign TA
1	4/19/22 8:03pm	AI course questions	How do I solve this?	Jack
2	4/19/22 8:03pm	AI course questions	How do I solve this?	Jack
3	4/19/22 8:03pm	AI course questions	How do I solve this?	Jack
4	4/19/22 8:03pm	AI course questions	How do I solve this?	Jack

Discussion Forum Analysis - Python Course



Student Teacher Topics

Current unresolved posts: 6

Today's average reply time (mins): 25

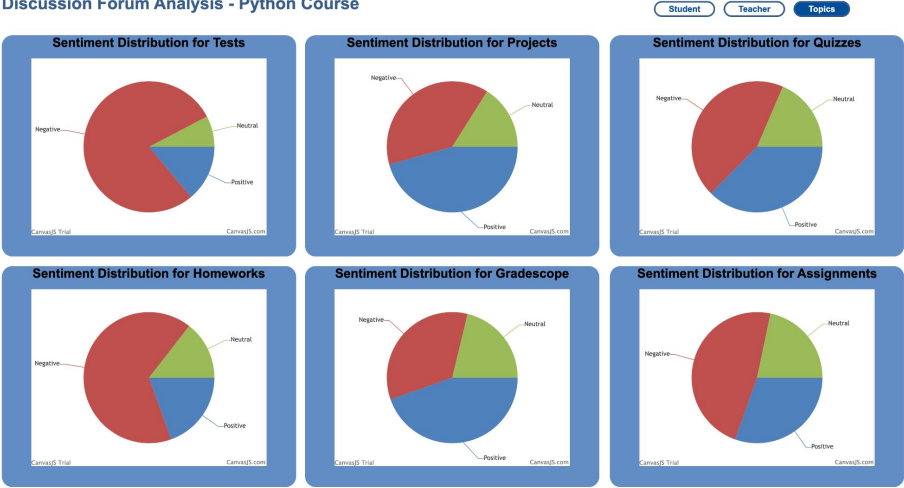
Current unresolved posts: 25

Unresolved Posts				
Post	Time	Subject	Question	Assign TA
737	4/19/22 8:03pm	Python course questions	How do I solve this?	Jack
737	4/19/22 8:03pm	Python course questions	How do I solve this?	Jack
737	4/19/22 8:03pm	Python course questions	How do I solve this?	Jack
737	4/19/22 8:03pm	Python course questions	How do I solve this?	Jack

Discussion Forum Analysis - AI Course



Discussion Forum Analysis - Python Course



Goals for this semester

- Extend dashboards to show data related to new courses
- Limit access to data to authorized users
 - professors of the course
- Make the dashboard responsive
 - so that it can be used on devices of all sizes

Goals for this semester (cont)

- Interface to navigate between different courses and view analysis
 - Experiment with different data representation
 - Further organize sentiment analysis by related categories
- Integrate user feedback in dashboard for ease-of-use
 - Work with research team to gain more user feedback or requested features
- Streamline backend to support more data and functionality

Conclusion

- Two dashboards for different subjects (Webdev)
 - Implement feedback from TAs after interviews
 - Research new visualizations needed for the new data
 - TA's and Teachers are good resources
- Data Science
 - Research different models and see how it helps answer research questions
 - Condense the data into a useful and concise frame that can be use to accurately display relevant information
 - Find how to make current Dashboard real time with future models

Citations

1. Mishra, Ankit, Sugasi Guatam, & Gehani Ritika. (2022). Web Development Team Report. *Georgia Tech Vertically Integrated Projects*.
2. “Topic Modeling: A Basic Introduction.” *Journal of Digital Humanities*, <http://journalofdigitalhumanities.org/2-1/topic-modeling-a-basic-introduction-by-megan-r-brett/>.
3. Yilmaz, Begum; “4 Machine Learning Use Cases in Sentiment Analysis” *AIMultiple*, <https://research.aimultiple.com/sentiment-analysis-machine-learning/>.

Thanks for listening!

Questions?